

in a **northerly** direction.

Q2(a)

At constant speed, reaction force = weight of man.

$$\therefore mg = 670 \text{ N}$$

$$\therefore m = \mathbf{68.3 \text{ kg.}}$$

2(b)

Between 2.0 s and 6.0 s, $R > mg$.

Since the man is slowing down or decelerating, he must be moving downward initially.

By N2L, $mg - R = m(-a)$ or $R > mg$

2(c)

At $t = 2.0$ s, the lift was moving with velocity u before decelerating and coming to a stop at $t = 6.0$ s By Impulse-Momentum Theorem applied to between 2.0 s and 6.0 s where there is a change of momentum and hence change in velocity,

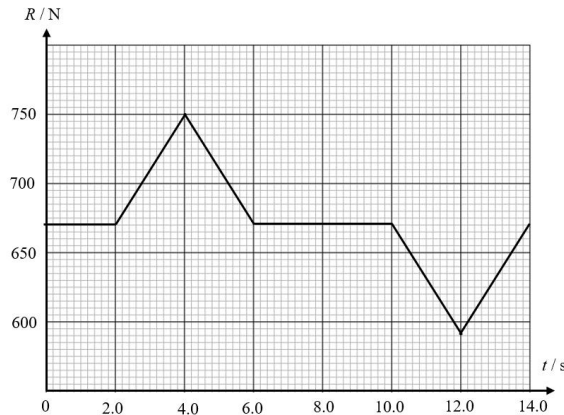
$$\Delta p = \int_2^6 F dt$$

$$68.3(0 - u) = \frac{1}{2} \times (6.0 - 2.0) \times (750 - 670)$$

$$\therefore u = -2.34 \text{ m s}^{-1}.$$

Hence, initially speed = **2.34 m s⁻¹**.

2(d)



2(e)

The total external force that the lift motor must counter is given by:

$$F_{\text{external}} = M_{\text{car}}g + M_{\text{load}}g - M_{\text{counterweight}}g$$

$$\therefore F_{\text{external}} = 1200 \times 9.81 + 420 \times 9.81 - 1320 \times 9.81 = 2943 \text{ N}$$

As the lift car is moving with constant speed,

$$\text{power exerted by lift motor} = P = F \times v$$

$$= 2943 \times 2.34 = \mathbf{6890 \text{ W.}}$$

Q3(a)